

SUMMARY

Software engineer with 2+ years building production backend and full-stack systems for Algeria’s largest classifieds marketplace, serving millions of users and requests (microservices, REST/GraphQL APIs, RabbitMQ, Docker, Kubernetes, Keycloak). Now building reliable agentic AI systems with human-in-the-loop controls, evaluation harnesses and observability. MSc Software Engineering student at the University of Innsbruck.

TECHNICAL SKILLS

Agentic AI & GenAI: LangGraph, multi-agent orchestration, tool calling, plan/verify loops, human-in-the-loop, RAG, hybrid search (BM25 + vector), reranking, embeddings, structured outputs, evaluation harnesses, prompt-injection defence, LLM APIs (OpenAI, Anthropic, Azure OpenAI)

Languages: Python, TypeScript, JavaScript, PHP, SQL, C/C++

Backend & Frontend: FastAPI, NestJS, Node.js, Laravel, REST, GraphQL, Microservices, RabbitMQ, React, Vue.js 3

Data: PostgreSQL, pgvector, Redis, MySQL, MongoDB, SQLAlchemy, Prisma ORM

DevOps & Security: Docker, Kubernetes, GitLab CI/CD, GitHub Actions, Linux, Git, Keycloak, OAuth 2.0 / OpenID Connect, JWT, RBAC

Observability & Tools: Sentry, OpenObserve, OpenTelemetry, Prometheus, Grafana, Jira, Agile/Scrum

AI ENGINEERING PROJECTS

RootCause – Multi-Agent SRE Assistant github.com/aziz-05/rootcause-agent
Python, LangGraph, FastAPI, PostgreSQL/pgvector, Redis, Docker, Kubernetes, OpenTelemetry

- Diagnoses real injected infrastructure faults (container kills, latency, pool exhaustion) via a plan → execute → verify agent loop.
- Human approval enforced in code for every write action; fixed tool registry and Kubernetes RBAC limit scope.
- 23-check CI gate on every pull request, including adversarial prompt-injection cases; 53 tests.

Souk Copilot – Production RAG & Analytics Platform github.com/aziz-05/souk-copilot
Python, FastAPI, PostgreSQL/pgvector, Redis, Docker, OpenAI / Azure OpenAI

- Routes questions to RAG or text-to-SQL, refusing rather than guessing when retrieval relevance is too low.
- Hybrid retrieval (BM25 + vector, rank fusion, reranking) with citations bound to source chunks.
- CI-gated evaluation (recall@k, MRR, groundedness); sandboxed read-only SQL; JWT row-level isolation; 103 tests.

PROFESSIONAL EXPERIENCE

Full Stack Web Developer | SARL Ouedkniss, Algiers, Algeria Nov 2023 – Present

- Built and maintained microservices and REST/GraphQL APIs (Laravel, NestJS, Node.js) behind React and Vue front-ends for a platform serving millions of users and requests, keeping core listing and search features fast and stable under heavy traffic.
- Worked directly with production databases (PostgreSQL, MySQL) and live data at scale, optimising slow queries and applying data changes safely, which improved response times while preserving data integrity.
- Implemented asynchronous, event-driven processing with RabbitMQ to decouple services and move heavy work off the request path, improving API responsiveness during traffic peaks.
- Secured user and service access with Keycloak (OAuth 2.0 / OpenID Connect) and role-based access control, centralising authentication across microservices.
- Containerised services with Docker and deployed them to Kubernetes through GitLab CI/CD pipelines with automated tests, making releases faster, repeatable and lower-risk.
- Integrated Sentry error tracking and OpenObserve logs and metrics, shortening the time to detect, diagnose and fix production incidents.
- Used AI coding agents for development, testing and code review, delivering Jira-tracked sprint work faster.

Network Engineer | Freelance, Algeria Mar 2020 – Nov 2021

- Designed, installed and configured LAN, WAN and WLAN networks from scratch for small businesses and startups, giving each client reliable, well-documented connectivity.
- Set up VPN and secure remote access so staff could work safely off-site, applying TCP/IP routing, switching and firewall configuration.
- Monitored network performance and diagnosed outages and slowdowns, resolving issues quickly to keep business operations running.
- Performed ongoing system and network maintenance (patches, service packs, security hardening), reducing vulnerabilities and unplanned downtime.

EDUCATION

MSc Software Engineering (in progress) | University of Innsbruck, Austria Oct 2026 – Present
Coursework: Machine Learning, Parallel Programming, Logic.

MSc Networking & Distributed Systems | University of Constantine 2 – Abdelhamid Mehri, Algeria 2019 – 2021

BSc Computer Science | University of Constantine 2 – Abdelhamid Mehri, Algeria 2016 – 2019
Coursework: distributed systems, microservices, network security, AI. Projects: EMODEC emotion API; QoS-AODV routing (C++).

LANGUAGES

Arabic (Native) | English (B2) | French (B2) | German (in progress)